QTC Module Report

Agnieszka Czeszumska
RCNP
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QTC converts input pulse into a NIM pulse with width proportional to the charge collected. The NIM pulse can then be recorded with a TDC avoiding an ADC. For the experimental setup, the goal is to replace the GR-DAQ consisting of the FERA-CAMAC system (20-30 µs/trigger deadtime) with the QTC module fed directly into the VME V1190 A. Theoretically, this could reduce the deadtime. This report summarizes tests of the QTC, its performance coupled with the plastic and $LaBr_3$ detectors and the VME, and describes optimization of parameters.

1. Overview of the QTC module

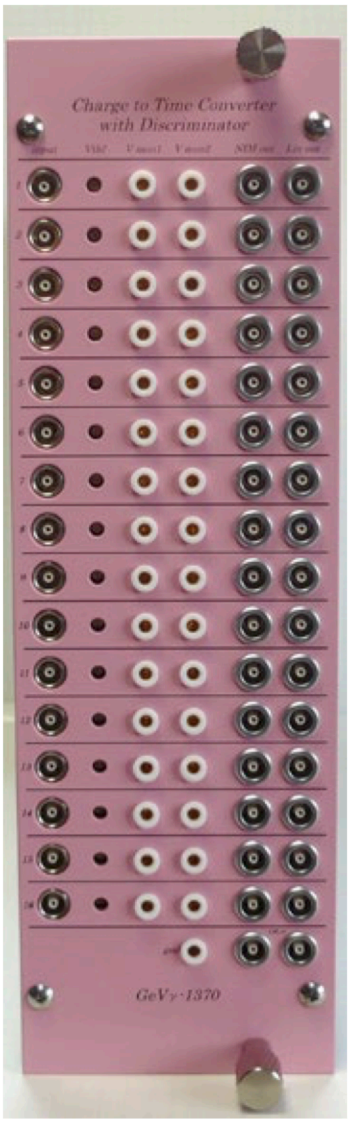


Figure 1 Front of the QTC module GeV g-1370 from Fuji Diamond

The GeV γ-1370 Fuji module accepts 16 input channels, and outputs NIM out, linear out, and an ECL output on the rear. The module requires a crate able to supply up to high currents, hence a mini crate is inappropriate to power the module. The power requirements are:

-6V: 3.8A (!)
+6V: 800mA
-12V: 40mA
+12V: 350mA

Importantly, care must be taken not to saturate the unit. The pulses have to be less than 1 µs, and a 10-ns width is preferable to simulate response of plastic detectors (a suitable pulser or a function generator can be used).

The QTC module has four adjustable settings: Vth1, Vth2, Integrate time, and PZC. The latter three are adjustable on the side panel, while Vth1 can be controlled from the front.

a) Vth1: threshold level for the input signal. Set it to below noise level.
b) Vth2: threshold level for the linear out output. Set it to above noise level.
c) PZC: Pole zero correction of the linear out output pulse.
d) Integrate time: modifies the linear out output and the length of the NIM output. It is important to optimize this parameter so it's long enough to accommodate the entire length of the pulse, but the setting it to the maximum integration time can degrade the performance of the module and reduces the resolution. An integration time that's too short clips the signal and results in missing pulses at larger amplitudes.

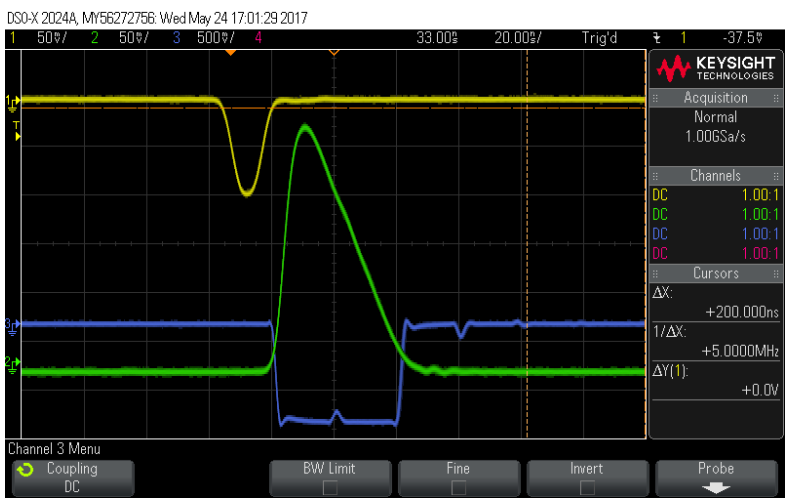


Figure 2: QTC Input (pulser, yellow) and output: NIM pulse (blue), linear out (green)

2. Summary of QTC Testing

a) MCA Tests

To test the basics of the module, we used a Kromec MCA unit, which recorded the "linear out" from the QTC module, and a function generator to produce an input signal. An Ortec pulser outputs pulses that are too long (~few microseconds) for the module to process. Good performance was found using pulses ~-100 mV in amplitude, and between 10 and 800 ns in

length. In general, it was found that output signal degrades (i.e. the resulting MCA histogram is no longer a single peak) if pulses are spaced less than 2 microseconds apart, because the linear out output takes time to reach the baseline. The delay through the module is about 10 ns.

To test the Integration Time (IT) variable, a 10-ns wide and -100-mV high pulse was fed into the QTC. Since the IT potentiometer does not have a readout, the IT is defined as the width of the NIM pulse produced by the "standard" pulse of 10-ns and -100 mV (see. Fig. 2 for reference). The linear out recorded by the MCA for three different IT's (40, 60, and 80 ns) is shown in Fig. 3. The pulses were fit to reveal that increasing IT increases the pulse width. The mean of the signal also increases non-linearly with increasing IT.

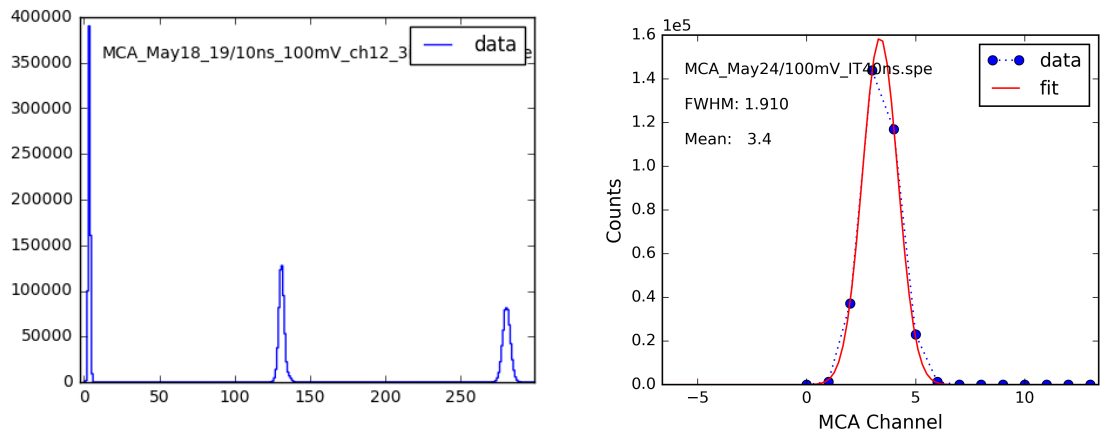
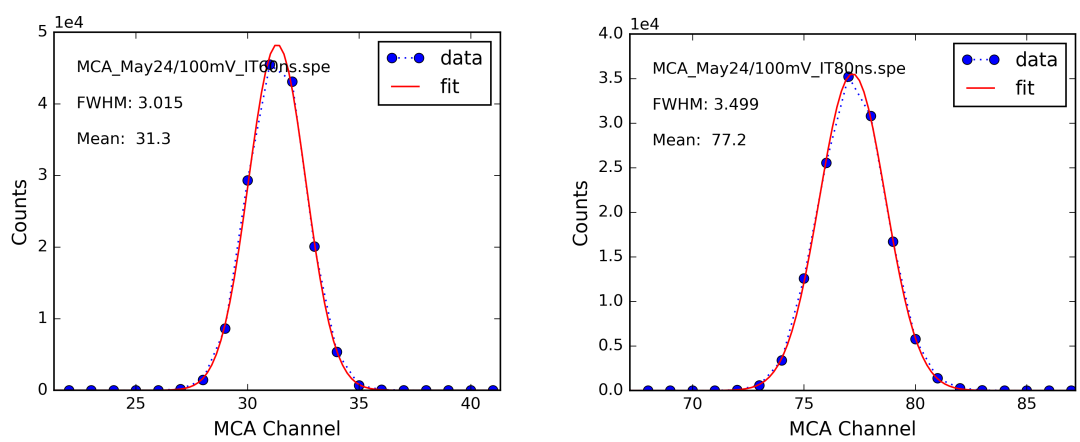


Figure 3 Linear out from three Integrate Time values



The range and saturation of the module was also studied by testing the MCA output with increasing pulse height. Fig. 4 shows the pulse voltage sweep (which was systematically increased in increments of 10 mV from -10 mV to -300 mV pulse height) for minimum IT value (left) and maximum IT (right). In both cases the signal saturates at higher pulse heights. The increase in peak with for the larger IT value is also apparent.

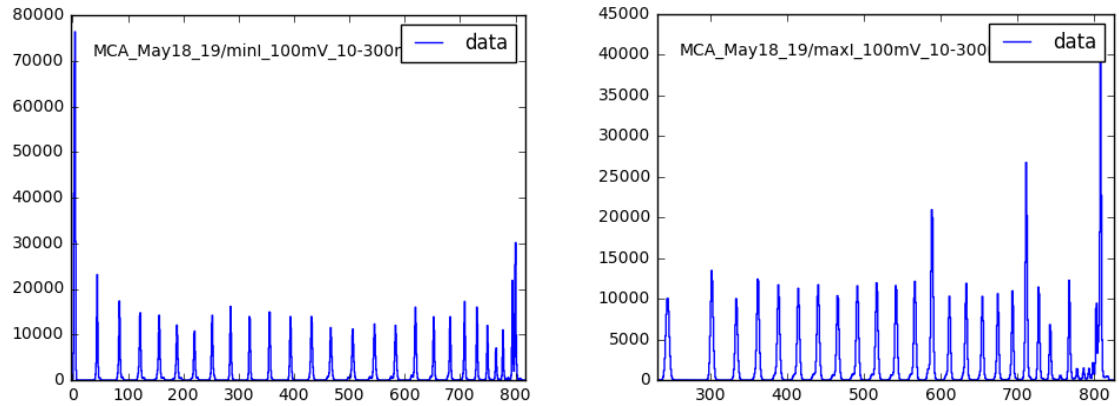


Figure 4 MCA linear out vs. various pulse heights

The general conclusion appears to be that for the 10-ns wide pulse, a 60-ns Integrate Time may be optimal. At the largest IT, the saturation may become a larger problem, as well as the loss of signal resolution.

b) TDC Tests

To test the setup in the experimental hall, the QTC module was placed in a VME crate and its NIM output was recorded with a TDC. The leading edge of the NIM pulse was recorded as the timing signal, and the length of the NIM pulse was processed as the signal amplitude. In parallel, the signal was recorded by the ADC as well. For the test signal, a function generator was used to produce a 10-ns-wide square pulse of varying amplitudes. To test the performance of the QTC against the real PMT signal, we connected the plastic scintillator, and recorded source data. Real data were collected during the CAGRA "Calibration Run" in May 2017 to ultimately test the QTC against the ADC under real experimental conditions, as well as in October 2017 using different settings, and lastly in December to test the QTC coupled with $LaBr_3$ detectors.

The pulser tests were useful for testing saturation and linearity of the module. In the voltage sweep in Fig. 5, the QTC output (y-axis) is plotted against ADC (x-axis) Saturation occurs around channel 1800 of the QTC. In following data taking, attenuators were used to keep the signal more proportional to the ADC and to prevent saturation. Attenuation of at least x5 is recommended. In Fig. 6, the same QTC vs. ADC response is plotted for fixed voltages. The relationship between the ADC and the QTC amplitude appears linear, as well as the QTC response vs. the input pulse height.

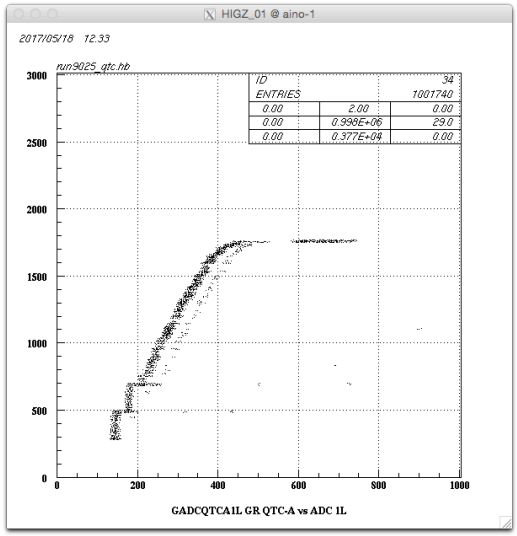


Figure 5 ADC (x-axis) vs. QTC (y-axis), pulser signal with increasing pulse height. Saturation of the QTC is apparent.

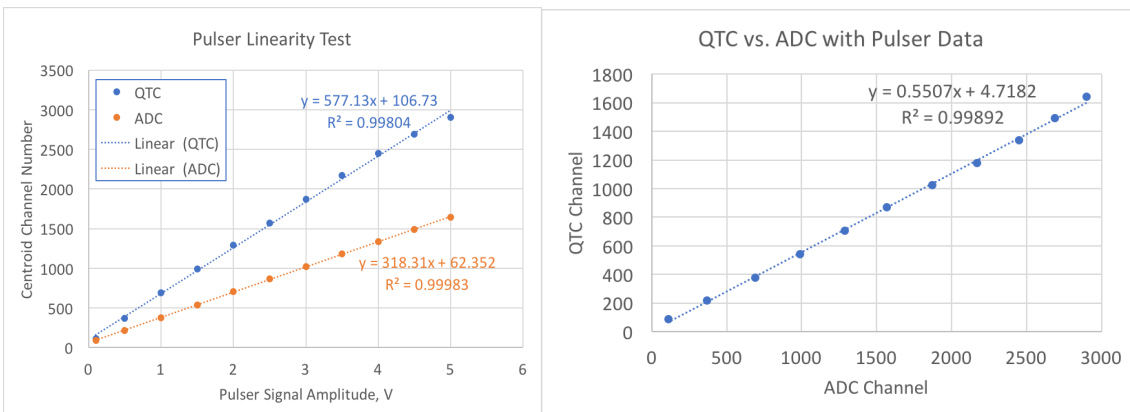


Figure 6 Left: Linearity of the QTC and ADC vs. pulser signal amplitude. Right: QTC vs. ADC response.

The pulser was also used to study signal loss at high count rates. It seems that above 500 kHz, the signal loss increases up to 10% (see Fig. 7). However, as will be discussed in the next section, this signal loss is also related to how the analyzer recorded the data. Various tests indicated that the VME system has 300 kHz limit.

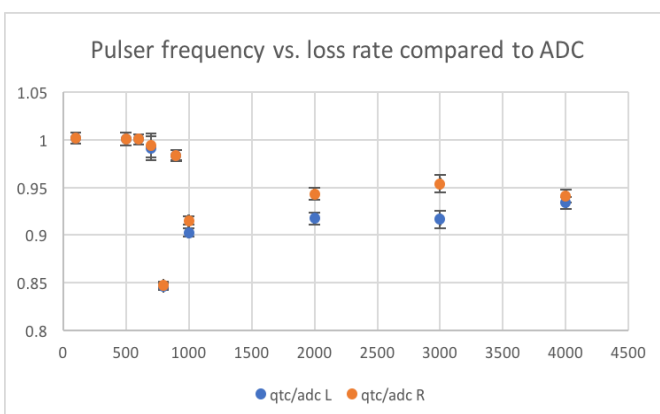


Figure 7 Signal loss (%) as a function of pulser frequency

3. Data Analysis and Results

This section summarizes results of analysis with the QTC coupled to the VME system, and compares it to the ADC. The analysis concentrates on the data collected for the Plastic scintillator and $LaBr_3$ detectors. The section starts with describing the analyzer and its role in interpreting the QTC data, and goes on to detail the performance of the QTC compared to the ADC, including signal loss, linearity, and noise.

a) Analyzer

The data parsed from the analyzer are the leading and trailing edges. Hence, the amplitude is deduced from: QTC_TRAILING[qtc_channel] – QTC_LEADING[qtc_channel], and the timing is set by QTC_TRAILING[qtc_channel] variable.

In the course of analyzing the QTC data (specifically Run1224 from October), it was uncovered that many events had an unphysical timing of trailing edge arriving before leading edge (see Fig. 8). It also became apparent that a single event may contain several edges, which the existing analyzer did not record, hence missing a large portion of pulses. In Fig. 9, the number of leading edges vs. trailing edges occurring in one event is plotted. Good events must have an even number of leading and trailing edges, with the leading edge proceeding the trailing edge.

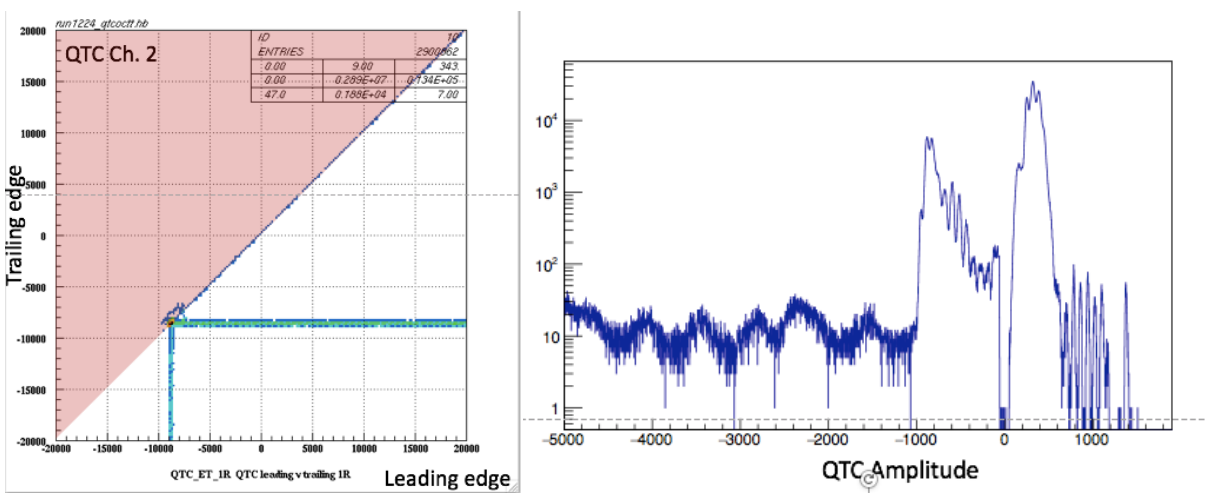


Figure 8 Unphysical QTC pulses (trailing edge comes before leading edge). Left: red region contains "good" events Right: data from ch. 0 of the QTC.

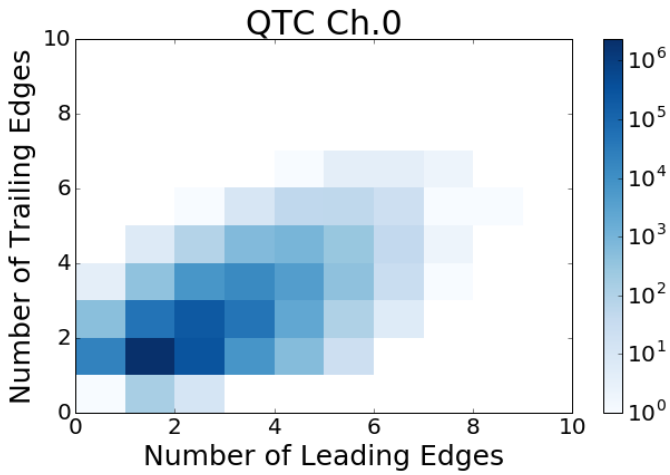


Figure 9 Run1224, QTC ch. 0 number of leading vs. trailing edges

The analyzer was therefore modified to correctly extract and recover multiple pulses in each event. The new analyzer is saved under fread_rcnp_new.c file. For each event, the leading and trailing edges are recorded for each channel into a 2D array: [QTC channel][edge number]. After the event is looped over for each channel and all the edges, the arrays are sorted from smallest to largest edge, then paired up correctly so that if trailing edge appears before the leading edge, it is skipped (or if two leading or trailing edges are in a row). If there is an error and an edge is skipped, or no edge pairs are detected, the QTC_ERRORS variable is incremented according to the specific error. The histogram values for QTC_ERRORS are:

0: L skipped
1: T skipped
2: no L+T pair

The new paired and correct edges are then saved into QTC_LEADARR_*nch[i]* and QTC_TRAILARR_*nch[i]* variables, where *nch* is the channel number and *i* is the edge number. So, the 1st leading edge for QTC channel 1 would be written as: QTC_LEADARR_1[0]. Channels start from 1, and edges start from 0.

In addition, the analyzer has a keyword for the QTC prompt timing gate, as the background can be substantial. The keywords are controlled by QTC_TSTART and QTC_END variables in the Histogram Maker. Initially they should be set to -20000 and 20000, since they may be different for each experimental setup.

When the prompt gate is applied, another histogram is created that contains the amplitude of the QTC (therefore, QTC_TRAILING – QTC_LEADING), summed over all the edges (hence contains data from all the multiple hits). The summed amplitude histogram is simply QTC_*nch* (i.e. QTC_1, QTC_2 etc). Here, the QTC_PAIRS_PROMPT counting the number of all valid pairs is already appended. The QTC_PAIRS variable counts all the valid pairs, regardless of the prompt gate.

The prompt gate becomes especially important when the background is very large, as is the case with the $LaBr_3$ data obtained in December, when the detectors were placed very close to the beamline. Fig. 10 shows the background reduction.

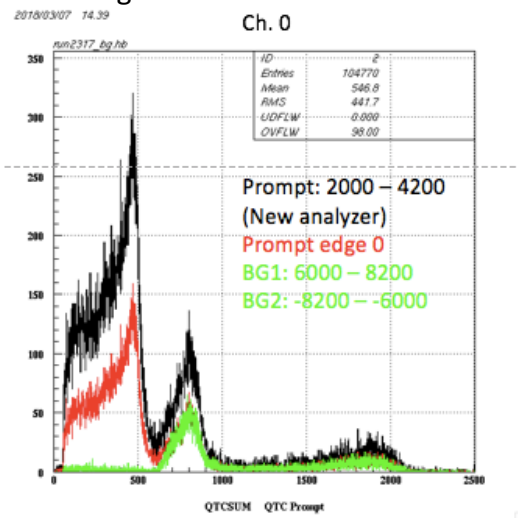


Figure 10 LaBr3 data with different timing gates applied.

Lastly, the number of edges allowed and the number of channels of the QTC read out is controlled by the max_edges and qtc_ch_max within the fread_rcnp.c file. It is currently set at 10 edges and 16 channels.

By running the new analyzer on the previously-analyzed data, nearly all missing counts were recovered. It also reduced the appearance of some strange features in the older data. Fig. 11 shows the old analyzer data from May run5067. The right channel had a smaller attenuation, and shows the reflection. Upper panes show the ADC and QTC amplitude signal, and the lower panes show the ADC signal without QTC coincidence (hence the missing QTC events), with ADC signal scaled and superimposed (gray). The peak around channel 700 corresponds to the QTC region where reflection occurs.

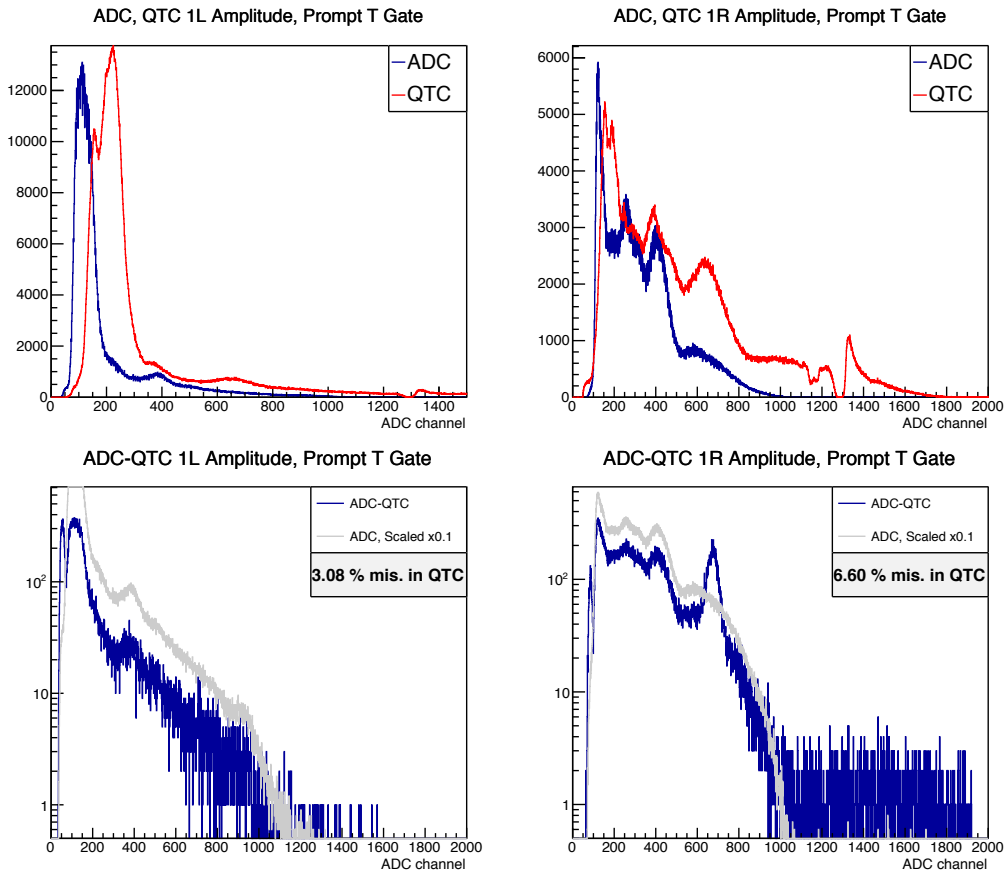


Figure 11 Old analyzer

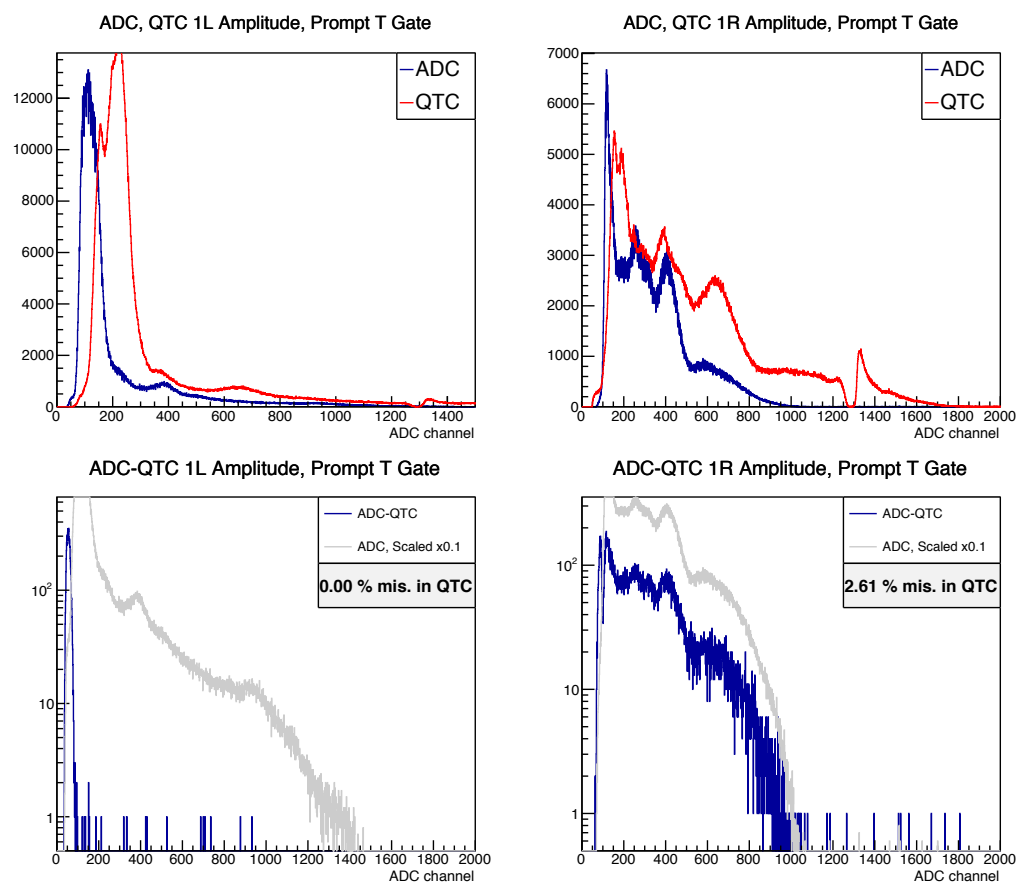


Figure 12 New analyzer

As can be seen in Fig. 12, the new analyzer recovers nearly all missing counts for 1L channel above the ADC pedestal when the GR_X coincidence is required (i.e. real events). For the 1R channel, the reflection may be at fault for the missing events, though they have been reduced by half, and are constant across the ADC amplitude (and also in coincidence with GR_X variable, and various others). The exact source of the missing counts appears to be the reflection but hasn't been completely understood. Linearity is not affected by the new analyzer.

b) Reflection

For the data from May, both channels (1L and 1R, i.e. ch. 0 and ch. 1) saw a reflection that was also visible in the scope signal. The reflection is visible as a dent in the data in Fig. 13. Its mechanism can be understood as follows: the plastic scintillator signal is split into two at the PMT. When the signal travels to the ADC, it is reflected back, and then is added into the signal that travels to the QTC, delayed by the cable length to and from the ADC. Thus, a positive reflection is introduced that cannot be easily removed, unless a single output is used, and the other output terminated. To avoid the reflection when the signal is split into two, attenuating the signal so that its amplitude appears below the reflection is recommended. The reflection should not be an issue if only one signal out is used, as confirmed with the oscilloscope.

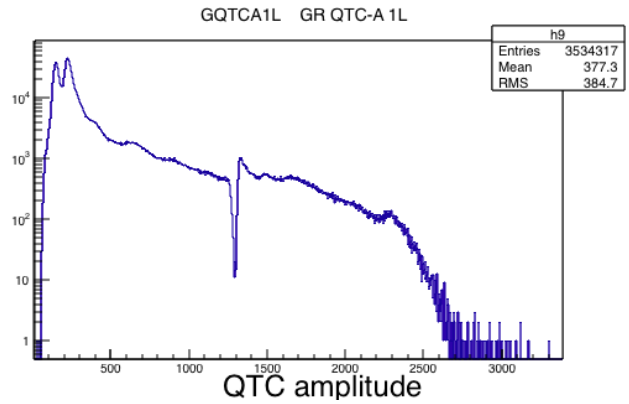


Figure 13 QTC amplitude for ch. 1, showing reflection as the dent around 1300.

c) Nonlinearity

By plotting the QTC vs. ADC, the linearity of the signals was studied. For the May runs, the linearity and its profile (mean along x-axis) is shown in Fig. 14.

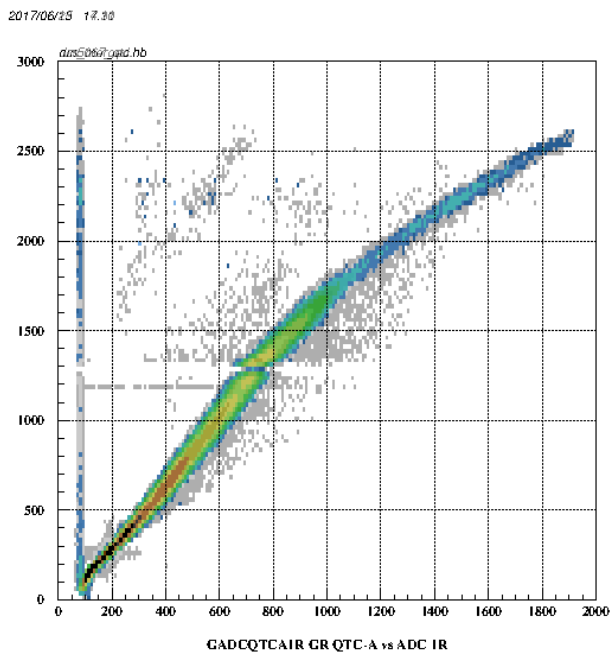


Figure 14 Linearity: ADC (x-axis) vs. QTC amplitude (y-axis), run 5067 in May.

Although the linearity seemed preserved for pulser runs, the real data shows the saturation of the QTC signal at higher amplitudes, but also some small deflections from linearity at lower amplitudes. However, when GR_X coincidence is required, most of the counts at higher amplitudes disappear.

The linearity from real data compared to the pulser runs is shown in Fig. 15. The source of the nonlinearity was not fully understood.

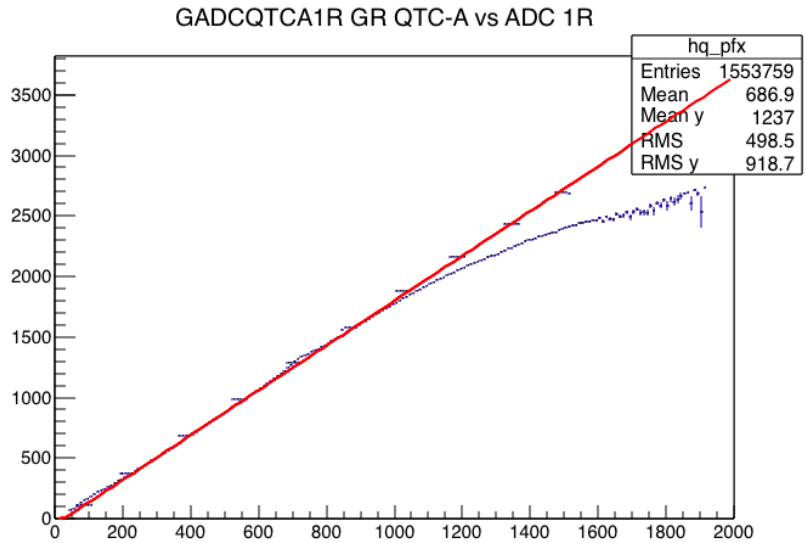


Figure 15 Linearity compared between data and pulser tests.

4. Conclusions

The QTC module was studied in detail, and this report only contains the most significant findings that reflect important points of operating the QTC module. First of all, it seems that the QTC module must have very fine-tuned parameters, i.e. for the case of using it with a plastic scintillator, the integration time should be set high, though care must be taken for the high integration time not to cause saturation of the signal or loss of resolution. The saturation should be taken care of by implementing attenuators. The signal also needs to be checked for reflection, and the analyzer implemented to handle multi-hits. However, even despite good settings, there may still be some missing counts and nonlinearity. The latter is especially important for the $LaBr_3$ detectors. As a conclusion, the QTC implementation should be carefully considered.